P03

Spectral Properties of the Graph Laplacian and Network Diffusion

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Domain DOM_MATH

Abstract

The graph Laplacian L = D - A acts as a discrete Laplace-Beltrami operator. Heat diffusion across network nodes du/dt = -L*u relaxes at rates dictated by the spectral gap and algebraic connectivity.

Abstract quoted from F. R. K. Chung et al., Spectral Graph Theory (AMS) (1997).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?